

8.5 Quiz: Rational Functions and Transformations — Markscheme

Version (a)

1. $f(x) = \frac{2}{x-3} + 1.$

(a) Vertical asymptote: $x = 3.$ A1

Horizontal asymptote: $y = 1.$ A1

(b) *“Describe fully” requires words, not symbols alone. One mark per correctly described transformation (A1A1A1).*

A vertical stretch (stretch / dilate / scale), scale factor 2 — direction (vertical) stated. A1

A translation (translate / shift / slide) 3 to the right — “right” explicit. A1

A translation 1 up (vertical direction may be omitted; assumed upward). A1

*Accept “factor of two”/“doubled vertically”; “parallel to the y-axis” for the stretch direction; “positive x-direction” for right. The word “units” is optional. **Reject “move”.***

Order: the stretch must precede the vertical translation; the horizontal translation may go in any position (deduct 1 mark if the vertical translation precedes the stretch).

Accept the two translations as a vector $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ for both translation marks, provided a translation word is used. A purely symbolic description with no transformation words (e.g. $\frac{1}{x} \rightarrow \frac{2}{x} \rightarrow \frac{2}{x-3} \rightarrow \frac{2}{x-3} + 1$) scores at most 1 of 3.

(c) $f(0) = \frac{2}{-3} + 1 = \frac{1}{3},$ so $(0, \frac{1}{3}).$ A1

Accept $y = \frac{1}{3}.$

(d) $\frac{2}{x-3} + 1 = 0,$ $\frac{2}{x-3} = -1,$ $x-3 = -2$ M1

$x = 1,$ so $(1, 0).$ A1

Accept $x = 1.$

2. $g(x) = \frac{2x+1}{x-1},$ $x \neq 1.$

(a) Vertical asymptote: $x = 1.$ A1

Horizontal asymptote: $y = 2$ (ratio of leading coefficients $= \frac{2}{1}$). A1

(b) x -intercept: $2x+1=0 \Rightarrow x = -\frac{1}{2},$ so $(-\frac{1}{2}, 0).$ A1

y -intercept: $g(0) = \frac{1}{-1} = -1,$ so $(0, -1).$ A1

Accept $x = -\frac{1}{2}$ and $y = -1$.

(c) Let $y = \frac{2x+1}{x-1}$ and swap x and y : $x = \frac{2y+1}{y-1}$ M1

$x(y-1) = 2y+1$, $xy-2y = x+1$, $y(x-2) = x+1$ A1

$g^{-1}(x) = \frac{x+1}{x-2}$ A1

(d) Domain of g^{-1} : $x \in \mathbb{R}$, $x \neq 2$. A1

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value of g from (a)).

Version (b)

1. $f(x) = \frac{3}{x+2} - 1$.

(a) Vertical asymptote: $x = -2$. A1

Horizontal asymptote: $y = -1$. A1

(b) “Describe fully” requires words, not symbols alone. One mark per correctly described transformation (A1A1A1).

A vertical stretch (stretch / dilate / scale), scale factor 3 — direction (vertical) stated. A1

A translation (translate / shift / slide) 2 to the left — “left” explicit. A1

A translation 1 **down** — “down” must be explicit (the upward default does not apply here). A1

Accept “factor of three”/“tripled vertically”; “parallel to the y -axis” for the stretch direction; “negative x -direction” for left. The word “units” is optional. **Reject “move”.**

Order: the stretch must precede the vertical translation; the horizontal translation may go in any position (deduct 1 mark if the vertical translation precedes the stretch).

Accept the two translations as a vector $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$ for both translation marks, provided a translation word is used. A purely symbolic description with no transformation words (e.g. $\frac{1}{x} \rightarrow \frac{3}{x} \rightarrow \frac{3}{x+2} \rightarrow \frac{3}{x+2} - 1$) scores at most 1 of 3.

(c) $f(0) = \frac{3}{2} - 1 = \frac{1}{2}$, so $(0, \frac{1}{2})$. A1

Accept $y = \frac{1}{2}$.

(d) $\frac{3}{x+2} - 1 = 0$, $\frac{3}{x+2} = 1$, $x+2 = 3$ M1

$x = 1$, so $(1, 0)$. **A1**

Accept $x = 1$.

2. $g(x) = \frac{3x - 2}{x + 1}$, $x \neq -1$.

(a) Vertical asymptote: $x = -1$. **A1**

Horizontal asymptote: $y = 3$ (ratio of leading coefficients $= \frac{3}{1}$). **A1**

(b) x -intercept: $3x - 2 = 0 \Rightarrow x = \frac{2}{3}$, so $(\frac{2}{3}, 0)$. **A1**

y -intercept: $g(0) = \frac{-2}{1} = -2$, so $(0, -2)$. **A1**

Accept $x = \frac{2}{3}$ and $y = -2$.

(c) Let $y = \frac{3x - 2}{x + 1}$ and swap x and y : $x = \frac{3y - 2}{y + 1}$ **M1**

$x(y + 1) = 3y - 2$, $xy - 3y = -x - 2$, $y(x - 3) = -(x + 2)$ **A1**

$g^{-1}(x) = -\frac{x + 2}{x - 3}$ **A1**

Accept the equivalent form $\frac{x + 2}{3 - x}$.

(d) Domain of g^{-1} : $x \in \mathbb{R}$, $x \neq 3$. **A1**

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value of g from (a)).

Version (c)

1. $f(x) = \frac{4}{x - 1} + 2$.

(a) Vertical asymptote: $x = 1$. **A1**

Horizontal asymptote: $y = 2$. **A1**

(b) “Describe fully” requires words, not symbols alone. One mark per correctly described transformation (A1A1A1).

A vertical stretch (stretch / dilate / scale), scale factor 4 — direction (vertical) stated. **A1**

A translation (translate / shift / slide) 1 to the right — “right” explicit. **A1**

A translation 2 up (vertical direction may be omitted; assumed upward). **A1**

Accept “factor of four”/“quadrupled vertically”; “parallel to the y-axis” for the stretch direction; “positive x-direction” for right. The word “units” is optional. **Reject “move”.**

Order: the stretch must precede the vertical translation; the horizontal translation may go in any position (deduct 1 mark if the vertical translation precedes the stretch).

Accept the two translations as a vector $(\frac{1}{2})$ for both translation marks, provided a translation word is used. A purely symbolic description with no transformation words (e.g. $\frac{1}{x} \rightarrow \frac{4}{x} \rightarrow \frac{4}{x-1} \rightarrow \frac{4}{x-1} + 2$) scores at most 1 of 3.

(c) $f(0) = \frac{4}{-1} + 2 = -2$, so $(0, -2)$. A1

Accept $y = -2$.

(d) $\frac{4}{x-1} + 2 = 0$, $\frac{4}{x-1} = -2$, $x - 1 = -2$ M1

$x = -1$, so $(-1, 0)$. A1

Accept $x = -1$.

2. $g(x) = \frac{4x+3}{x-2}$, $x \neq 2$.

(a) Vertical asymptote: $x = 2$. A1

Horizontal asymptote: $y = 4$ (ratio of leading coefficients $= \frac{4}{1}$). A1

(b) x-intercept: $4x + 3 = 0 \Rightarrow x = -\frac{3}{4}$, so $(-\frac{3}{4}, 0)$. A1

y-intercept: $g(0) = \frac{3}{-2} = -\frac{3}{2}$, so $(0, -\frac{3}{2})$. A1

Accept $x = -\frac{3}{4}$ and $y = -\frac{3}{2}$.

(c) Let $y = \frac{4x+3}{x-2}$ and swap x and y : $x = \frac{4y+3}{y-2}$ M1

$x(y-2) = 4y+3$, $xy - 4y = 2x+3$, $y(x-4) = 2x+3$ A1

$g^{-1}(x) = \frac{2x+3}{x-4}$ A1

(d) Domain of g^{-1} : $x \in \mathbb{R}$, $x \neq 4$. A1

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value of g from (a)).